Data Engineer Gap-Bridging Roadmap (12–16 Weeks)

1■■ Orchestration & Workflow Management (Weeks 1–3)

Gap: Employers expect Airflow/Prefect for scheduling & monitoring.

Action: Learn Apache Airflow basics (DAGs, operators, scheduling, logging). Mini-project: Daily pipeline that pulls CSV → transforms with Python → loads into Snowflake. Stretch goal: Add task retries + email alerts.

2■■ Big Data Processing with PySpark (Weeks 2–6)

Gap: You've done ETL in Talend & SQL, but AU jobs demand distributed data processing.

Action: Learn PySpark (DataFrames, joins, aggregations, window functions). Project: Process a 1M+ row dataset. Do cleaning, aggregation, push output to Snowflake. Document performance gains.

3■■ Streaming Data Pipelines (Weeks 5–8)

Gap: Many senior roles want real-time experience (Kafka/Spark Streaming).

Action: Learn Kafka basics (producer, consumer, topics). Project: Real-time order ingestion → Kafka → Spark Streaming → Snowflake. Optional: Tableau dashboard for near real-time updates.

4■■ Advanced Cloud Data Engineering (Weeks 7–11)

Gap: You know AWS basics, but need deep, production-level knowledge.

Action: Build end-to-end AWS data lake project: Ingest raw data → S3, Transform with PySpark on EMR, Store curated data → Redshift/Snowflake, Orchestrate with Airflow. Learn IAM, cost optimization, monitoring. Bonus: Try Terraform or AWS CDK for infra deployment.

5■■ Data Transformation (dbt) + Governance (Weeks 10–13)

Gap: Modern AU companies use dbt for ELT & want governance experience.

Action: Learn dbt core (models, tests, docs). Build transformations in Snowflake with dbt. Add tests for data quality + documentation.

6■■ Capstone Project & Portfolio (Weeks 12–16)

Gap: Need one flagship project that screams 'production-ready data engineer'.

Action: Capstone project: E-commerce Analytics Platform. Batch ETL (Airflow + PySpark → Snowflake), Streaming (Kafka → Snowflake), dbt models (dimensional modeling, tests), BI dashboard (Tableau/Power BI). Deliverables: GitHub repo + diagrams + README with business case.

■ Rules While Learning

- Weekly time split: 10 hrs learning, 10 hrs project building.
- Weekly deliverable: Push code + notes to GitHub.
- Bi-weekly update: Refine resume with new skills/projects.
- Monthly checkpoint: Mock interviews (SQL, Python, system design).

■ Recommended Resources

- Airflow: Astronomer Academy, official docs.
- PySpark: Databricks Academy, Spark by Examples (YouTube).
- Kafka: Confluent Kafka 101 (free).
- AWS Data Engineer: A Cloud Guru, AWS Skill Builder.
- dbt: dbt Learn (free).
- Interview Practice: LeetCode (SQL, Python), StrataScratch (SQL business cases).